r/DefendingAIArt

Honest Question About AI Water Consumption

Posted by u/Arrestedsolid • • 0 points • 32 comments

Hi everyone. I wanted to ask an honest question about the whole “AI consumes a ton of water” argument that gets brought up a lot by anti-AI folks.

I’ve seen some common counterpoints already, like the fact that there are many other industries we rely on daily that consume far more water, and that AI data centers aren’t unique in that sense. But I’m still curious about the core of the claim itself.

Is AI’s water usage actually a real, meaningful environmental problem? And if it is, how big of a concern is it in practice compared to other existing technologies and infrastructures? Or is this argument mostly just grasping at straws to have something to complain about AI?

I’m genuinely trying to understand where the reality sits here. Thanks in advance.

Comments

u/IndigoFenix • 2 points • 2026-01-05 08:48 UTC

The water thing is a local/municipality issue, not a global/environmental/ecological one.

Some (I repeat, **SOME**) AI companies have been using water from communities that have water issues. This is something that will likely be regulated in the near future because it directly affects local voters. People who try to lump it in with things like climate change are doing so *purely* as an attempt to rope in the ignorant.

There is a small uptick in *energy* usage, but it's barely even worth mentioning against things like transportation, and considering that a fair chunk of the actual usage of AI is focused on making these things more efficient or less necessary, it's likely to result in a global net positive overall.

u/Exotic-Plankton6266 • 2 points • 2026-01-03 17:28 UTC

Power and water consumption is tough to calculate not only because of the nature of computing tech (power consumption is not a constant and changes on the fly in computers), but also because there's so many variables and middlemen spread around the world to track. To be honest I'm not sure _any_ of these water consumption studies are accurate. This is why they say it takes between 14 to 22 burgers produced to consume as much water as the average person. That's a 150% difference between the lower and higher bound.

I think the bigger question is could something be done about it and yes, it could. It's a matter of policy, not tech. If you forced all new datacenters (or any other business water consumer) to follow new policies they would complain for a bit but then get with the program just like we do when new laws are announced. Because what else are they going to do, not make datacenters? But that cuts into profits so that would never happen. Let them rampage through the countryside, it's the only way line goes up.

u/markaction • 3 points • 2026-01-03 14:14 UTC

Maybe an engineer can add insight to my question -- but if the water cooling system was a just a closed-loop, and the water is recycled -- just like a water-cooled CPU in your PC, is this problem now solved?

So it is just people building these data-centers on the cheap, and we should just demand they pay a little more to have a closed-loop for cooling?

> **u/squirtnforcertain** • 2 points • 2026-01-03 21:34 UTC
>
> Yeah but antis would probably count each "lap" a gallon makes and count it again anyways.

u/TheRealCorwii • 3 points • 2026-01-03 13:50 UTC

I mean, I'm literally running local AI on my laptop that takes 0 water to run. Laptops and water definitely can't mix. So how can I be generating videos here at home without consuming a single fluid ounce of water? Cause it runs on my Nvidia RTX card, not my bathtub.

> **u/squirtnforcertain** • 1 points • 2026-01-03 21:32 UTC
>
> Well you ARE consuming water via your electricity usage, but agree it's not as egregious as antis make it out to be.

u/CapitanM • 3 points • 2026-01-03 11:58 UTC

You think about La Liga.

Between 4 and 30 cameras recording in 4K, sending uncompressed footage. Processing all that and sending it to thousands and thousands of people.

That's just the video transmission, not counting all the encoders and security measures that may be involved.

Each game consumes as much energy as if each of its viewers were using artificial energy hundreds of times over.

u/Content-Audience252 • 2 points • 2026-01-03 10:26 UTC

Guys, data centers use a ton of water that we can never get back! And it’s not like we have 332 million cubic miles of the stuff lying around…

u/WestGotIt1967 • 3 points • 2026-01-03 10:02 UTC

I think it is about 5% of the problem.

It is a super easy target for lazy activists.

ReL activists would be metaphorically burning down the national association of manufacturers and shut down massive industry from the USA to China to the EU. No kid alive today that is doing virtue signal performative action has the gumption to go that far.

Like Connery said in the untouchables:
If you open the ball on these people you better be prepared to go all the way. BecUse they won't stop until one of you is dead.

You wanna get Capone? Here's how you get him. He pulls a knife, you pull a gun. He sends one of yours to the hospital, you send one of his to the morgue! That's the Chicago way, and that's how you get Capone!

u/Popular-Hornet-6294 • 2 points • 2026-01-03 08:57 UTC

I'm deliberately creating AI images to destroy humanity. And if the water argument is fake, I'm just creating pretty pictures.

u/arbitrary_student • 0 points • 2026-01-03 07:46 UTC

It uses as much water (and electricity) as any data center does. That makes its impact location-dependent.

The biggest thing to note is that AI is only about \~20% of total data center use in the world, and data centers are only about 1% of total electricity use - and a way, way, way, way, way smaller % of water usage. So basically it's only an issue in the sense that humanity is consuming way too much stuff in general, which has basically nothing to do with AI specifically.

Copy-pasting from my own previous comment (can't directly link to reddit in this sub):

One of the key points of this comic is that the whole AI thing is a distraction from the actual problem, which is the rampant use of resources by billionaires and society in general.

AI accounts for something like 0.1% of total electricity consumed in the US. Water usage is similarly negligible as a total. (edit: changed link) https://www.carbonbrief.org/ai-five-charts-that-put-data-centre-energy-use-and-emissions-into-context/

That study that came out recently that said AI has caused the same CO2 emissions as New York city should have been an eye-opener in *the other direction*. Know what else emits as much CO2 as New York city? New York city. **One single city** produces as much pollution as all of AI does. Add the rest of the USA, and then all the other cities & countries in the world, and suddenly it seems silly to focus on something consuming 1 city's-worth of resources. Corporations & billionaires *love it* when AI gets blamed for everything.

It's not that AI doesn't use a lot of resources, it's that it's a drop in the bucket. At worst, it's the straw that may break the camel's back. Electricity and water consumption were already skyrocketing without AI, and we're struggling with these as a society because of the insane resource use of *everything* - from cars, to computing in general (e.g. the data centers for social media), agriculture, plastic still being in every product, pretty much anything you can think of. It's not just AI, it's *everything*. If you can name it, we're overusing it.

AI is largely irrelevant on its own, and this comic is pointing out that it's just a distraction from the fact that, even if AI didn't exist, we'd be in the exact same situation. Deleting AI now would not solve our problems, and we need to focus on what would: resolving the billionaire crisis.

More info (edit)

Someone pointed out I pasted the wrong source link. That's fixed now, and I may as well drop more info here as well.

As an exercise, let's estimate some things based on the ["as much energy as New York"](https://www.theguardian.com/technology/2025/dec/18/2025-ai-boom-huge-co2-emissions-use-water-research-finds) figure (it's actually carbon emissions, but it's still a good proxy). New York uses approx. [5 billion kilowatt hours](https://www.nyc.gov/content/climate/pages/energy-infrastructure) of electricity per year, and the *entire US* uses [about 4 trillion](https://www.eia.gov/energyexplained/electricity/use-of-electricity.php).

If we do a bit of math on that we find that *all AI globally* is consuming about 0.125% of *just the USA's yearly total*. This is probably an undershoot, so we'll move on from this - but it does highlight the ballpark of the figures we're working with.

For direct figures, below is a relevant excerpt [from the first link above.](https://www.carbonbrief.org/ai-five-charts-that-put-data-centre-energy-use-and-emissions-into-context/)

> [...] data centres are currently responsible for just over 1% of global electricity demand and 0.5% of CO2 emissions, according to IEA data.

> As it stands, AI has been responsible for around 5-15% of data-centre power use in recent years [*excluding crypto-currency mining*], but this could increase to 35-50% by 2030, according to another report prepared for the IEA.

> However, the 530TWh rise in electricity demand in data centres by 2030 would only be 8% of the overall increase in demand that the IEA projects.

> This is less than electric vehicles (838TWh) or air conditioning (651TWh). It is considerably less than the 1,936TWh growth expected in industrial sectors by 2030.

In short, at the moment the world is on around 1% of total electricity being used by *all data centers*, and if we take the highest current estimate from the report, around 15% of that is AI. That means AI is using about 0.15% of electricity globally. I've seen other estimates that range from 15-25% in 2025, so it may be as high as 0.25% or so.

Per the point of my original comment about energy & resource consumption, if all AI was immediately destroyed it would accomplish approximately **fuck all**.

Here are a couple of studies with lots of numbers for further reading.

https://www.cell.com/patterns/fulltext/S2666-3899(25)00278-8

https://www.iea-4e.org/wp-content/uploads/2025/05/Data-Centre-Energy-Use-Critical-Review-of-Models-and-Results.pdf

u/[deleted] • 1 points • 2026-01-03 07:43 UTC

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> **u/AutoModerator** • 1 points • 2026-01-03 07:43 UTC
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u/Whilpin • 8 points • 2026-01-03 07:34 UTC

519ml per GPT 5 prompt. (roughly)

Sounds like a lot right? Thats not **just** the datacenter though, that's inclusive of the power required (turns out power plants need water for cooling too)

However when you compare it to... say...

A single hamburger patty... It's less than a drop in the bucket.

A single hamburger patty would be roughly equivalent to 600 gallons of water. So it'd take 4 prompts (roughly) = 1 gallon. works out to 1 hamburger == ~240,000 prompts.

> **u/[deleted]** • 1 points • 2026-01-03 10:28 UTC
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> > **u/AutoModerator** • 2 points • 2026-01-03 10:28 UTC
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u/jksdustin • 4 points • 2026-01-03 07:33 UTC

Not so much speaking about online and subscription based models, but you can use comfyUI for free on your home PC if you have a half decent GPU, and it takes less energy than playing a videogame with no water use (unless you use it for cooling)

> **u/xoexohexox** • 1 points • 2026-01-03 18:06 UTC
>
> I use water to cool my PC but it's closed loop, I don't have to top it off. The water gets pumped through a radiator cooled by fans and then back into the block.

u/One_Fuel3733 • 8 points • 2026-01-03 07:17 UTC

A pretty decent writeup with stats and sources and all that good stuff

https://andymasley.substack.com/p/the-ai-water-issue-is-fake